The Islamia University of Bahawalpur.

Department of Information Technology

Project proposal

Title:

AI Expression Detection(Website)

Version1.0

By

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SCOPEDOCUMENTREVSIONHISTORY

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Title:

AI Expression Detection

Project Category: (Selectallthemajordomainsofproposedproject**)**

○ **A-**DesktopApplication/InformationSystem ● **B-**WebApplication/WebApplicationbasedInformationSystem
● **C-**ProblemSolvingandArtificial Intelligence ○ **D-**SimulationandModeling ○ **E-**SmartphoneApplication ○ **F-**SmartphoneGame ○ **G-**Networks ○ **H-**ImageProcessing ○Other(specifycategory)_____

Description

This project involves the development of aro bust facial exression and demographic detection system using the deep face library, Integrate advanced machine learning model stop redict emotions, age, gender, and race from uploaded image built with Djang other back end and a user friendly-interface, the system aims to address accuracy and usability challenges in existing solution.

Introduction

Demographic and facial expression identification have been adopted across sectors like healthcare, education, security and marketing. These types of systems analyze facial features in order to predict a person's emotions, age, gender, and race which aids in the analysis of customers' behaviours or the adaptation of a product/service to meet individual needs. However, most existing systems have common limitations including: poor accuracy levels, insufficient functionality, and difficulty with diverse data sets. In order to overcome the current limitations in facial expression and demographic identification systems, this project will be creating a facial detection system in Django that utilizes the Deep Face library; which is the industry's leading company that provides all support tools necessary for developing face detection applications. A user-friendly web interface will be available to allow customers to upload images to receive complete analysis reports related to their images. The proposed facial detection system will combine existing face recognition models such as Arc Face and Retina Face which will enhance the performance of prediction when provided with difficult input data sets for predicting age, gender, race and emotion. The structure of the system will be adjustable allowing for future features such as: real-time face analysis or API integration. The goal of the proposed facial detection system is to be a better version of existing systems and to provide customers with an effective and accessible facial detection system.

ProblemStatement

What problemdoesyour software solve?
Why youare developingthissystem?
Does thesamesystemalreadyexist?
If yes, how will are-implementationaidyourlearning?

What skillsdoyouexpecttolearnfromthisproject? (Usually in 14-16 sentences)

Problem Solution for Proposed System

What problem does your soft ware solve?

The software I am developing addresses the issue of low accuracy and limited functionality that many of today's emotion/age/gender/race detection systems have. Many of these systems do not perform well at detecting emotion due to differences in age, skin tone, etc., as well as not being user-friendly. The software I am developing will be able to detect people's emotions and age(s) accurately with the help of advanced tools such as Deep Face and Retina Face.

Why you are developing this system?

I am designing a system that allows users to easily detect both facial expressions and demographics. My design should make it easier and quicker for people to get accurate readings.

This project's purpose is not only to develop a method (the facial expression and demographic detection system) but also to understand how AI technology and the Internet could be used to address actual issues.

Does the same system already exist? If yes, how will are implementation aid your learning?

While systems such as VGG-Face and FaceNet have been developed to provide face recognition capabilities, they are often highly specialised, focusing on only one type of feature, which makes them difficult to use for many users. Therefore, I will be able to build these systems and learn how to improve them while still providing them to users in a practical way. It will also allow me to address challenges such as dealing with low-quality images and making the software more efficient and usable by reducing processing time and improving user experience.

Whats kills do you expectto learn from this project?

By this project I will be able to gain knowledge of working with advanced AI models, understanding ways to effectively utilise these advanced AI models, developing skills in creating web applications with Django, and efficiently managing databases.

I will also gain knowledge of integrating AI into real-world applications, improving systems based on end-user input.

The overall goal of this project is to educate me about how to design and develop entire solutions using AI, starting from idea generation through to implementation.

Related System Analysis/Literature Review

Discuss existing/similar systems relevant to your proposed project. At least provide three examples of existing systems and give brief explanations on how they relate to your proposed project.

The literature review of research-based project will provide sufficient data in place of related systems analysing; you must cite examples from peer-reviewed journals.

Table1 Related System Analysis with proposed project solution

ApplicationName	**Weakness**	**ProposedProject Solution**
VGG-FaceModel	Computationally intensive;limitedto facial recognition.	Combines lightweight, optimized models like Arc Face for multi-attributedetectionwith improved efficiency.
Face net	Limited to facial verification;struggles with lighting and occlusion.	Enhances robustness using Retina Face for better face detection in varied conditions and supportsmulti-attribute analysis.
OpenCV-Based Systems	Limited analysis of expressions/demographics; lower accuracy.	Integrates Deep Face modelsforhighaccuracy andcomprehensivefacial attribute detection.

Advantages/Benefits of Proposed System

High Accuracy: Utilizes advanced models like Arc Face and VGG-Face for precise predictions.
Versatile Analysis: Detects multiple attributes (emotion, age, gender, race) simultaneously.
User-Friendly Interface: Simple web interface for easy image uploads and result viewing.
RobustPerformance: Retina Face ensures accurate face detection in challenging conditions.
Scalability: Modular design allows easy future expansions and real-time analysis.
Real-TimeProcessing: Potential forreal-time image and video analysis in future versions.
Accessibility: Web-based platform accessible tousers without technical expertise.

Scope

The aim of the project is to produce a machine learning system capable of detecting and predicting emotions, age, gender and race for uploaded images through sophisticated ML models. A web interface will allow users to upload images and receive the results. The backend for the system will be in Django and use it to process images, send and receive user requests, and interact with the Deep Face Library to provide predictions to the user. The overall architecture of the system is going to be modular, meaning that future development can include real-time analysis of videos as well as integration with other external API's. There will be a database module within the system that will maintain user data and history of analysis. The system is designed to be scalable; meaning it will accommodate an increasing amount of data and future expansion and upgrading. The system is designed with the future in mind, including online model training and multiple image types.

Modules:

Write down the modules of the proposed project. Don't forget to mention special/new features. Briefly explain your one module in 6 to 8 sentences.

Explanation of a Module: Module is a section of a program that performs a task. Programs consist of modules, each of which contains one or more routines. The term routine is synonymous with procedure, function, and subroutine.

Example:

Enterprisere source planning(ERP)software-iscomprisedofseverallargemodules(forexample,finance, supply chain and payroll, etc.), which may be implemented with little or no customization.
(Brieflyexplain eachmodule withrespectto major functionalityin user context)

Module1: Front end Module:

This part of the system lets users upload their images and see the analysis results. It provides an easy-to-use interface for interaction.

Module2: Back end Module:

This module handles the processing of the uploaded images. It communicates with the Deep Face library to make predictions about emotions, age, gender, and race based on the images.

Module3: Database Module:

This module stores the results of the analysis and keeps track of user data, including history of the images processed and their corresponding results.

Module4: Training Module:

This feature allows the system to be updated by training the model with new data. This helps the systembecome more accurate and adaptable over time as it learns from more examples.

System Limitations/Constraints

Dependency on Pre-TrainedModels:

The system relies on pre-trained models like Deep Face, limiting flexibility for customizations.

Performance on Low-Quality Images:

Accuracy may drop for images with poor resolution, extreme lighting, orocclusions.

Hardware Requirements:

High computational power (e.g., GPUs) may be needed for efficient processing, especially with large datasets.

SoftwareProcessMethodology

1. **Planning Phase:** Define objectives, breakdown tasks, and plan sprints.
2. **Design Phase:** Design system architecture and user interface.
3. **Development Phase:** Develop system in small increments, test in after each sprint.
4. **Testing Phase:** Test modules, fix bugs, and refine features.
5. **Deployment Phase:** Deploy the system to a server and monitor performance.
6. **Maintenance Phase:** Collect feedback, fix issues, and add new features.

Tools and Technologies

Mention all the hardware/software tools and technologies with version number which will be used in implementation of the project. Write about the APIs, language(s), SDK(s) etc. which you will use for implementation.

Example:

Table2ToolsandTechnologiesforProposedProject

	Tools	Rationale
Tools And Technologies	PyCharm	IDE
	SQLite	DBMS
	Django	Webframeworkfordeveloping the backend.
	MSWord	Documentation
	MSPower Point	Presentation
	Technology	**Rationale**
	Html	Structureandlayoutofweb pages.
	CSS	Styling and design of web pages.
	JavaScript	Adding interactivity and dynamicfeaturestowebpages.
	Bootstrap	Responsivedesignframework for web pages.
	Python	Backend processing andlogic.
	Deep Face Library	Pre-trainedfacial recognition anddemographic detection models.
	Retina FaceDetector	Advanced face detectiontool.

Conclusion

The project's combination of machine learning models and a web-based interface provides an effective solution for facial expression and demographic classification. Predicting emotions, ages, genders, and races is possible through the combined use of the Deep Face Library and Django Framework. Technical and non-technical users will both find that the user-friendly interface of the system makes it very accessible. The modular construction also allows for future expansion and/or improvement of the tool such as real-time video analysis and integration with an API. Although there are some issues; such as reliance on pre-trained models, hardware requirements, etc. All of those challenges can bring about an improvement to the overall accuracy and usability of the systems currently available. Future progress can lead to more uses in security, education, health care, and marketing where there will be an interest in using the project to evaluate facial features from the provided data set. The project's combination of machine learning models and a web-based interface provides an effective solution for facial expression and demographic classification. Predicting emotions, ages, genders, and races is possible through the combined use of the Deep Face Library and Django Framework. Technical and non-technical users will both

find that the user-friendly interface of the system makes it very accessible. The modular construction also allows for future expansion and/or improvement of the tool such as real-time video analysis and integration with an API. Although there are some issues; such as reliance on pre-trained models, hardware requirements, etc. All of those challenges can bring about an improvement to the overall accuracy and usability of the systems currently available. Future progress can lead to more uses in security, education, health care, and marketing where there will be an interest in using the project to evaluate facial features from the provided data set.

References:

DeepFaceLibraryDocumentation:
https://github.com/serengil/deepfaceOfficialdocumentationfortheDeepFacelibrary,whichprovides pre-trained models for facial recognition and analysis.

DjangoFrameworkDocumentation:
https://docs.djangoproject.com/OfficialDjangodocumentationforwebapplicationdevelopment, including backend setup, routing, and database management.

ResearchArticlesonFacialRecognitionandMachineLearning:
Researchpapersandjournalsonvariousmachinelearningmodelslike VGG-Face,ArcFace,andRetina Face, which provide insights into facial analysis techniques and their applications.

BootstrapDocumentation:
https://getbootstrap.com/docs/OfficialdocumentationfortheBootstrapframework,whichisusedfor responsive web design.

PythonDocumentation:
https://docs.python.org/OfficialPythondocumentationforunderstandinglanguagefeaturesandlibraries used in backend development.